RINGER'S LACTATE

Balanced crystalloid

Not with ceftriaxone or blood

CHEMICAL NAME Compound sodium lactate (Hartmann's): Na^+ 131, K^+ 5, Ca^{2+} 2, Cl^- 111, lactate 29 mmol/L; ~278 mOsm/L

CLASS Balanced isotonic crystalloid (Hartmann's solution)

TRADE NAMES Hartmann's solution · Lactated Ringer's (LR) · Compound sodium lactate

THERAPEUTIC USES / INDICATIONS

- Fluid resuscitation (trauma, burns, sepsis, surgery) — balanced crystalloid preferred.
- Peri-operative fluid replacement.
- Replacement of GI losses (diarrhoea), burns (Parkland formula).
- DKA (balanced alternative to saline).

ADMINISTRATION

- **IV** peripheral or central line.
- Do not co-administer with ceftriaxone (calcium precipitate) or in the same line as blood products (calcium → clotting of citrated blood).
- Bolus 250–500 mL with reassessment.

DOSAGES

- **Resuscitation:** 500 mL bolus over <15 min (children 10–20 mL/kg), reassess.
- **Burns (Parkland):** 4 mL × kg × %TBSA over 24 h — half in first 8 h from time of burn.
- **Maintenance:** as prescribed (contains less Na^+ than plasma).

PHARMACOKINETICS

- Distributes in extracellular fluid.
- Lactate metabolised by the liver to bicarbonate → buffering effect (impaired in severe liver failure).

PHARMACODYNAMICS

- Restores **extracellular volume** with electrolyte composition close to plasma → less hyperchloraemic acidosis than 0.9% saline.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phlebitis.
- Oedema with excess.

ADVERSE EFFECTS (SERIOUS)

- Fluid overload.
- Hyponatraemia (slightly hypotonic) — caution in head injury/raised ICP.
- Hyperkalaemia (minor K^+ content; generally safe even in renal failure).
- Metabolic alkalosis with large volumes; raised lactate reading (not true lactic acidosis).

CONTRAINDICATIONS & CAUTIONS

- Fluid overload; hypercalcaemia; severe hyperkalaemia.
- Concurrent ceftriaxone (especially neonates); same line as blood.
- **Caution:** severe liver failure (lactate metabolism), raised ICP (slightly hypotonic).

KEY INTERACTIONS

- **Ceftriaxone:** precipitation; contraindicated in neonates; flush lines in others.
- **Blood products:** clotting — separate line.
- **Digoxin:** calcium content.

NURSING MANAGEMENT

- Fluid balance, vital signs, urine output, signs of overload.
- Burns: calculate from time of injury; adjust to urine output (0.5–1 mL/kg/h adults).
- Check compatibility with drugs and blood.
- Monitor U&E.

PATIENT EDUCATION

- Report breathlessness or swelling, or pain at the drip site.

KEY POINTS

- Balanced crystalloid (Hartmann's).
- Not with ceftriaxone or blood in same line.
- Resuscitation, surgery, burns (Parkland).
- Lactate → bicarbonate in liver.

SOURCES CHECKED **[960]** Hartmann's Solution Compound Sodium Lactate Intravenous Infusion BP (UK SmPC) **[381]** Evans et al., Intensive Care Med 2021

Full references p. 432–457